

Week 3

3.1 | Business Understanding



⌚ 30 minutes

The Team Data Science Process (TDSP) is an agile and iterative data science methodology aimed at delivering predictive analytics solutions and intelligent applications efficiently. It is designed to improve team collaboration and learning by suggesting how team roles work best together. TDSP includes best practices and structures from Microsoft and other industry leaders to help towards the successful implementation of data science initiatives, with the goal of helping companies fully realise the benefits of their analytics program. The process provides a generic description that can be implemented with different kinds of tools and includes key components such as a data science lifecycle definition, a standardised project structure, infrastructure and resources recommended for data science projects, and tools and utilities recommended for project execution.

The data science lifecycle outlined by TDSP (see the following figure) follows stages: Business Understanding, Data Acquisition and Understanding, Modelling, and Deployment, leading to the final stage, Customer acceptance. The process also recommends a standardised project structure, infrastructure and resources for data science projects, and tools and utilities for project execution. It provides an initial set of tools and scripts to jump-start the adoption of TDSP within a team and outlines the key personnel roles and their associated tasks for a data science team that standardises on this process.

Data Science Lifecycle

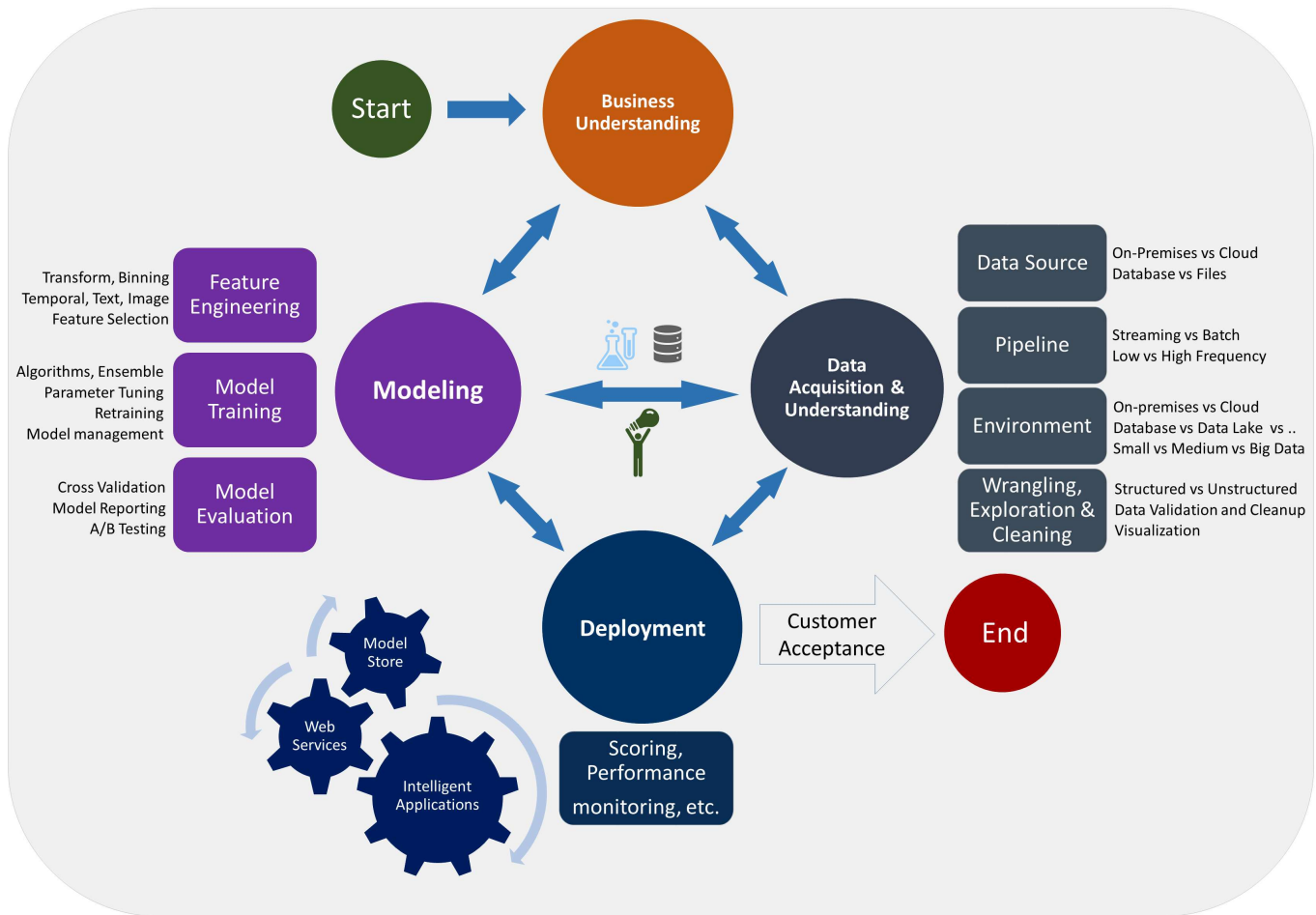


Figure 1: Data Science Lifecycle. Source: <https://learn.microsoft.com/en-us/azure/architecture/data-science-process/overview>

The challenge

You will build your understanding of the business implications of the churn prediction project so that you can apply this to the fictional company ConnectTel.

The output



Completion of the 'Business Understanding' section in the 'ConnectTel Project Documentation Template'.

The task

In this task, you will review the business context for the fictional company ConnectTel, then use a dataset to predict customer churn. You will play the role of project manager.

***ConnectTel is different from the Modern Furniture Co. case study used in the Assessment 1. The challenges related to the ConnectTel case study will help you familiar with the process and requirements for Assessment 1 around Modern Furniture Co.**

Business context

You have been tasked with a data science project for a telecom company called ConnectTel. In recent times, a new competitor has entered the market, offering an enticing plan to prospective customers. The telecom company is concerned about the potential loss of its existing customers to this new competitor.

In the absence of a ready dataset specific to this project, the project lead proposes utilising open data sources to fulfil the project's objectives. Open data sets related to telecom customer churn are readily available and can be used for analysis.

The primary goal of this project is to predict customer churn and identify the key factors that contribute to a customer's decision to switch to a different telecom provider. To achieve this, we will leverage publicly available telecom customer data.

The project will culminate in a comprehensive report designed to be easily comprehensible by a diverse audience. This report will encompass the motivation behind the analysis, the steps undertaken, the findings derived from the analysis, and the conclusions drawn from the findings.

Project conversations

Review the following conversations to gain a deeper understanding of the project context and your role as the project manager. We will follow through data scientist Emily's recommendations in the upcoming challenges this week.

Exploring datasets

John Anderson (CEO of ConnectTel): With the emergence of aggressive competition in the market, there's a pressing need for us to safeguard our customer base from being lured away. The new plans offered by our competitors are attractive, and we must act decisively to retain our customers. What's our strategy to counter this effectively?

Emily Davis (Data Scientist at ConnectTel): John, your concerns are absolutely valid. To address this challenge, I recommend that before diving into our internal data, we first explore an open dataset in the telecom industry. These datasets often contain a wealth of anonymised customer information, including behaviour patterns and service preferences, which are reflective of broader market trends. Utilising this open dataset will allow us to benchmark our services and customer satisfaction against industry standards. This initial analysis can provide us with valuable insights into churn predictors and customer preferences, which we can then compare and enhance with our proprietary data.

You (Project Manager at ConnectTel): That's an interesting approach. Could you elaborate on how we can integrate these open datasets with our internal data for a comprehensive analysis?

Emily Davis (Data Scientist): Certainly. The open dataset can offer us a broad perspective on customer behaviour in the telecom sector. Key data points include demographic information, service usage patterns, customer satisfaction indices, and churn rates. By analysing these factors, we can identify industry-wide trends and customer expectations. Following this, we can align this external data with our rich internal dataset, encompassing customer demographics, service preferences, billing details, and customer experience metrics. This two-pronged approach will not only provide us with a comparative analysis but also help us identify unique opportunities for ConnectTel to differentiate itself in the market and reduce churn effectively.

John Anderson (CEO): This sounds like a well-rounded strategy, using external benchmarks to complement our internal data. It will give us a clearer picture of where we stand in the market and how we can leverage our strengths. Let's proceed with this plan and utilise these insights for strategic customer retention.

Emily Davis (Data Scientist): Absolutely, John. By harnessing both external and internal data sources, we can gain a comprehensive understanding of the market dynamics and

customer expectations. This approach will enable us to develop more targeted and effective strategies to keep ConnectTel competitive and maintain our commitment to excellent customer service.

Reflect:

What is an open dataset? Why are the team interested in the open dataset instead of their real customer data?

The Team Data Science Process lifecycle (TDSP)

Review the following readings to understand the requirements in the business understanding stage of TDSP:

- [What is the Team Data Science Process?](#)  (Azure Machine Learning n.d.)
- [The business understanding stage of the Team Data Science Process lifecycle](#)  (Azure 2024)

Instructions

Review the task materials, then:

1. Download and review the following dataset (e.g., what fields/columns are included in the datasets, how many records/rows are included). You will use this dataset in the next two challenges this week:



Telecom customer churn dataset (XLSX 1.4MB)



2. Download the ConnectTel Project Documentation Template.



ConnectTel Project Documentation Template (DOCX 37KB)

3. Complete the Business Understanding' section. You can do extra research (e.g., industry reports, scientific articles) to provide appropriate details in this section (e.g., how AI and data science are used within the Telecom industry to predict churns, what are the business values and outcomes of customer churn prediction, what data commonly is used for customer churn prediction).



Come prepared to share what you have learned about Customer Churn Prediction. Start planning for the business understanding session for the Modern Furniture Co. case study.

Supporting resources

- Huang, Bingquan, Mohand Tahar Kechadi, and Brian Buckley. 'Customer churn prediction in telecommunications.' Expert Systems with Applications 39.1 (2012): 1414-1425. Go to the **Reading List** in the navigation bar on the left side of your screen to access this reading.
- [Case Study: Churn Prediction](#) (Inside Bloomberg, 2018).